

Python

Constant: A constant is an identifier whose value cannot be changed throughout the execution of a program whereas the variable value keeps on changing.

There are no constant in python, the way they exist in C and Java.

In Python, It is not possible to define constant whose value cannot be changed.

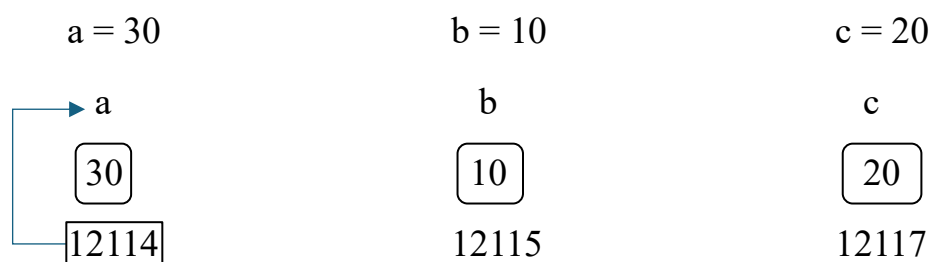
In Python, Constants are usually define on a module level and written in all capital letters with underscores separating words but remember its value can be changed.

Ex:- PI

TOTAL

MIN_VALUE

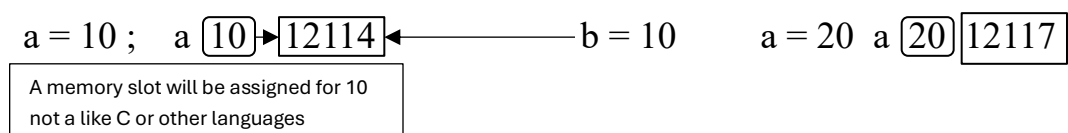
Variable: In C ,Java or some other languages, a variable is an identifier or a name, connected to memory location.



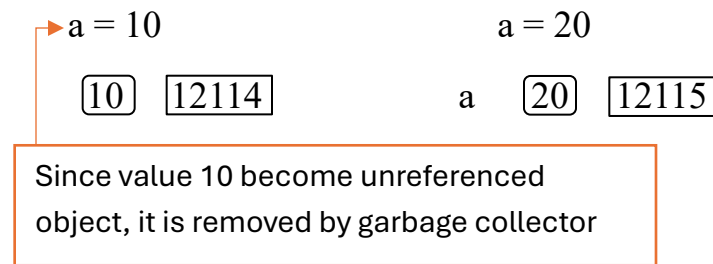
In C or other languages, a slot is generated in memory just for a or b or c.

But in Python, a slot is generated in memory just for the value of the variable.

In Python, a variable is considered as tag that is tried to some value. Python considers value as objects.



In Python memory slot generate just for the value not for the variable.



Rules:

- 1) Every variable name should start with alphabets or underscore (_).
- 2) No spaces are allowed in variable declaration.
- 3) Except underscore no other special symbol are allowed in the middle of the variable declaration
- 4) A variable is written with a combination of letters, numbers and special characters (_).
- 5) No Reserved Keywords.

Example:

Do

A
a
name
name15
_city
Full_name
FullName

Don't

and (Reserved Keyword)
15name
#city
Full\$Name
Full Name

In Python IDE:

```
>>>a = 10
>>>a
10
>>>id(a)
140706393187248
>>> b = 10
>>>b
10
>>>id(b)
140706393187248
```

Data Type: Datatype represents the type of data stored into a variable or memory.

Types of data type:-

- 1) Built-in Data type
- 2) User define Data type

In Python, we don't define datatype.

1) Built in Data Type:

These datatypes are provided by Python Language.

- 1) None Type
- 2) Numeric Types
- 3) Sequences
- 4) Sets
- 5) Mappings

None Type: None datatype represents an object that doesn't contain any value.

Numeric Type / Number:

Following are the Numeric Data Type:-

Int

Float

Complex

Int – The int datatype represents an integer number. An integer number without any decimal point or fraction part. In Python, It is possible to store very large integer number as there is no limit for the size of an int datatype.

Ex: 20, 10, -50, -1002

y = 10

pin_code = 564512 (this is an integer data type, because the value is integer)

To check the datatype:

```
>>>type(y)
```

```
<class 'int'>
```

Float – The float data type represents floating point numbers. A floating point number is a number that contain a decimal point.

Ex: 25.25, 10.5, -46.58

price= 25.25

run_rate=-0.8

value=5.1e5

Complex – A complex number is a number that is written in the form of $a + bj$ or $a + bJ$. Where,

a = Real part of the number

b = Imaginary part of the number

j or J = Square root of -1

a and b may contain integer or float number.

Ex:

$5 + 7j$, $0.8 + 2j$

`com=5+7j` (this is a complex datatype variable, just because of the complex number.

Bool Type:

The bool datatype represents Boolean value True or False. Python internally represents True as 1 and False as 0.

Ex: True, False

`True + True = 2`

`True - False = 1`

2) User Define Data Type:

1) Array

2) Class

3) Module